Filters:

- One of the most important circuit building blocks in electronic engineering.
- Indispensible for communication systems and in most instrumentation systems.

Here,

(i) Try to understand the basic properties of 4 major filters

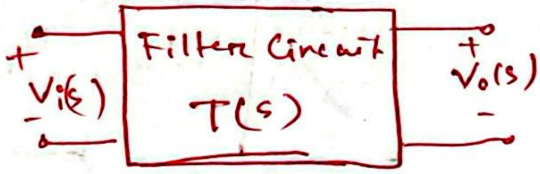
(ii) Learn the design and construction of filter circuits.

(iii) Drawing the bode plot

- A two port network,

  $V_i(s) \rightarrow$ input

  $V_o(s) \rightarrow$ output.



- Ratio of V_o to V_i taken as a function of frequency is the Transfer function, $T(s)$

$$T(s) = \frac{V_o(s)}{V_i(s)} \qquad s = j\omega \rightarrow \text{complex freq}^n.$$

Types:

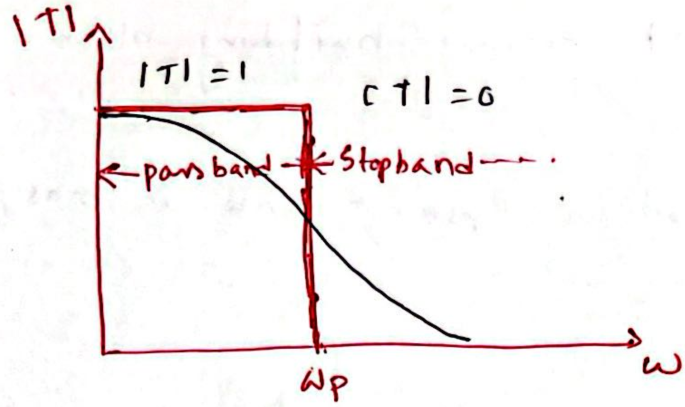
→ Low pass filter → allows low freq^n signal to pass through.

→ High Pass filter → allows high freq^n signal to pass through

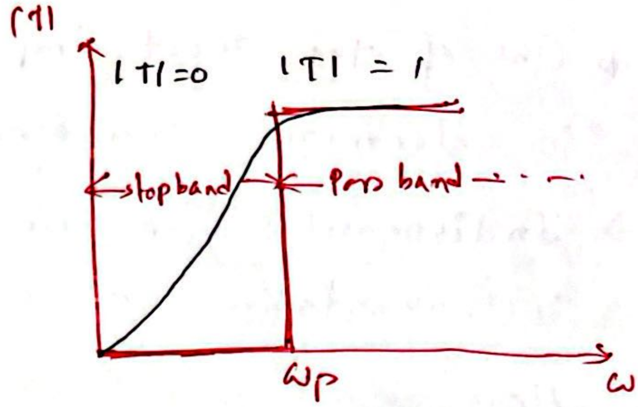
→ Band pass filter → allows a certain band of freq^n.

→ Band stop filter → blocks signal within a certain band of freq^n.

Ideal Filter Characteristic.

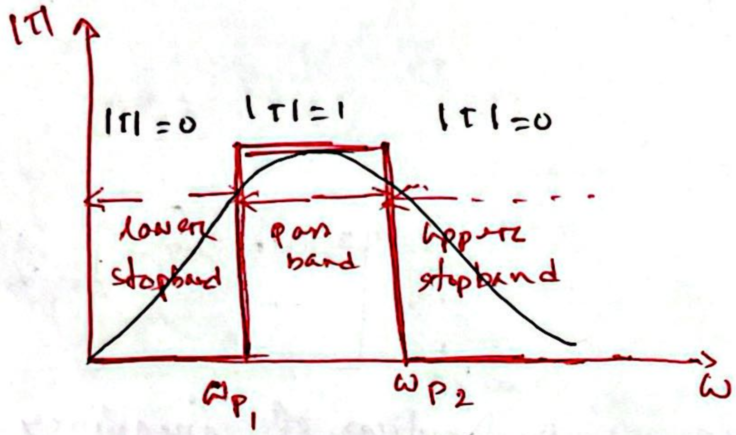


Low pass

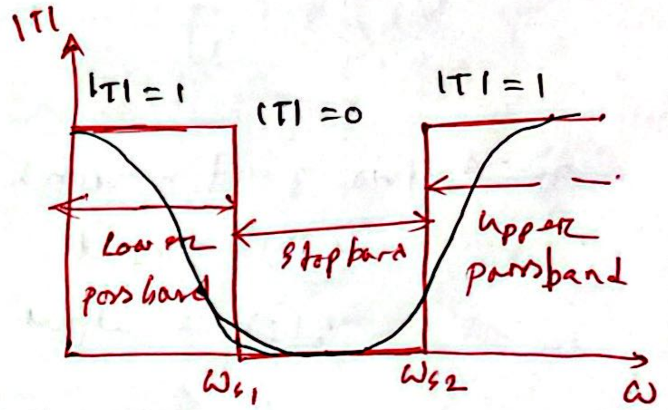


High pass

$\omega_p \rightarrow$ cut off freq



Band pass



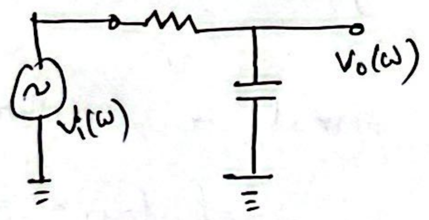
~~Sto~~ Bandstop

$\omega_{p_1}, \omega_{s_1} \longrightarrow$ lower cut off freq (ω_L)

$\omega_{p_2}, \omega_{s_2} \longleftrightarrow$ higher cut off freq (ω_H)

Passive RC filters:

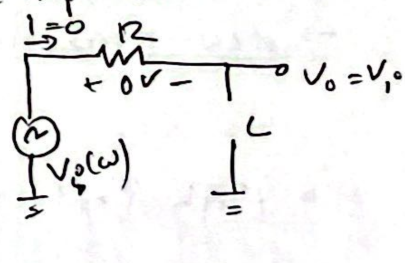
Use passive electrical components.

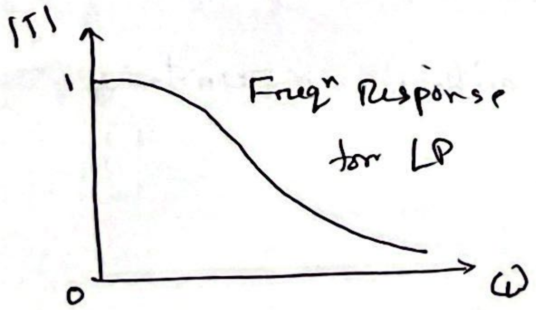


At, $\omega = 0$

$$X_c = \frac{1}{\omega c} = \alpha$$

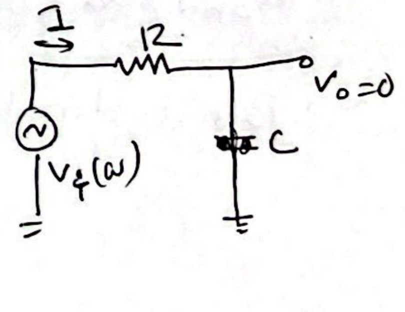
$$T = \frac{V_o}{V_i} = 1$$



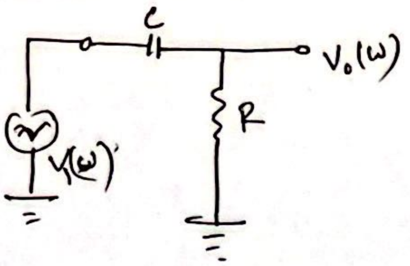


At $\omega = \alpha$

$$X_c = \frac{1}{\omega c} = 0$$



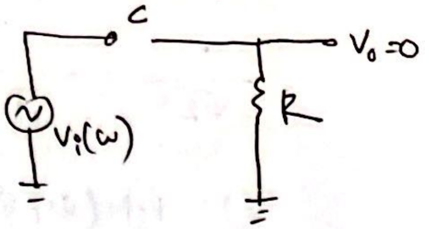
2)

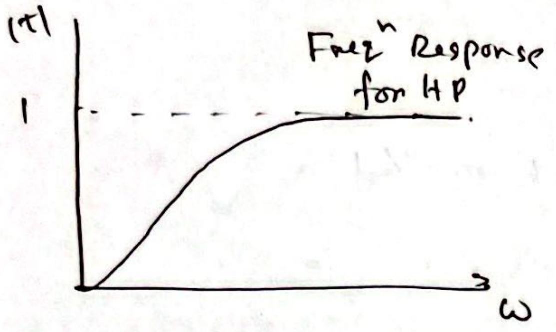


At, $\omega = 0$

$$X_c = \frac{1}{\omega c} = \infty$$

$$T = \frac{V_o}{V_i} = 0$$

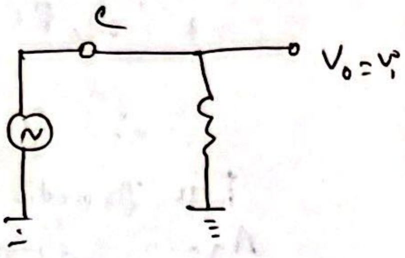




At, $\omega = \infty$

$$X_c = \frac{1}{\omega c} = 0$$

$$T = \frac{V_o}{V_i} = 1$$



Transfer function:

HP

$$V_o = \frac{R}{R + 1/j\omega C} V_i$$

$$= \frac{1}{1 + 1/j\omega RC} V_i$$

$$T(\omega) = \frac{V_o}{V_i} = \frac{1}{1 + 1/j\omega RC}$$

$$|T(\omega)| = \left| \frac{1}{1 + 1/j\omega RC} \right|$$

$$= \frac{1}{|1 + 1/j\omega RC|}$$

$$= \frac{1}{\sqrt{1 + (1/\omega RC)^2}} = 0, \omega = 0$$

$$= 1, \omega = \infty$$

LP

$$V_o = \frac{1/j\omega C}{R + 1/j\omega C} V_i$$

$$= \frac{1}{1 + j\omega RC} V_i$$

$$\therefore T(j\omega) = \frac{V_o}{V_i} = \frac{1}{1 + j\omega RC}$$

$$\therefore |T(j\omega)| = \left| \frac{1}{1 + j\omega RC} \right|$$

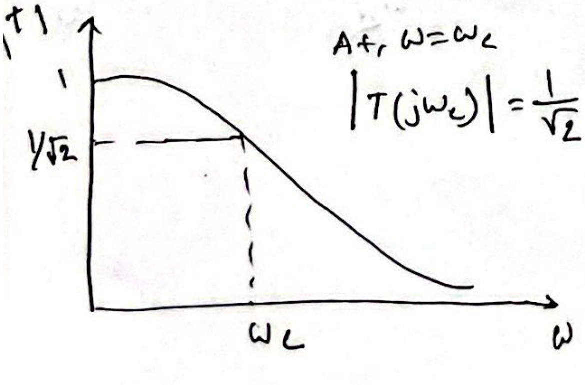
$$= \frac{1}{|1 + j\omega RC|}$$

$$= \frac{1}{\sqrt{1 + (\omega RC)^2}} = 1, \omega = 0$$

$$= 0, \omega = \infty$$

Cut off freq^n:

For lowpass:



At, $\omega = \omega_c$

$$|T(j\omega_c)| = \frac{1}{\sqrt{2}}$$

$$T(j\omega) = \frac{V_o}{V_i} = \frac{1}{1 + j\omega RC}$$

$$\therefore |T(j\omega)| = \frac{1}{|1 + j\omega RC|}$$

$$\therefore |T(j\omega_c)| = \frac{1}{|1 + j\omega_c RC|} = \sqrt{2}$$

$$\therefore \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{1+(\omega_c RC)^2}}$$

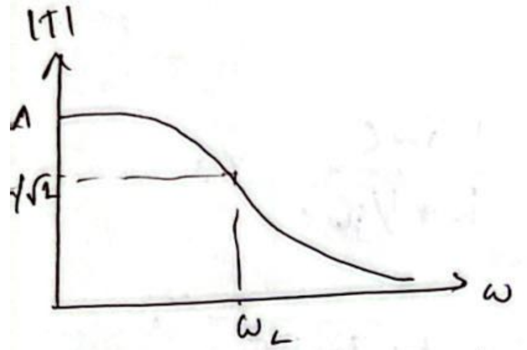
$$\Rightarrow 1+(\omega_c RC)^2 = 2$$

$$\Rightarrow (\omega_c RC)^2 = 1$$

$$\therefore \omega_c = \frac{1}{RC}$$

For practical case: gain can be >1 or <1.

Again, $T(j\omega) = \frac{V_o}{V_i}$



$$= \frac{1}{1+j\omega RC}$$

$$= \frac{1}{1+j\omega/\omega_c}, \left[\omega_c = 1/RC\right]$$

$$\therefore T(s) = \frac{1}{1+s/\omega_c}$$

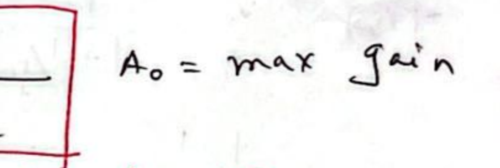
$$\boxed{\therefore T(s) = \frac{A_o}{1+s/\omega_c}} \quad A_o = \text{max gain}$$

General form of TF for LP

For HP:

$$T(j\omega) = \frac{V_o}{V_i} = \frac{1}{1+1/j\omega RC}$$

$$= \frac{1}{1+\omega_c/j\omega}$$

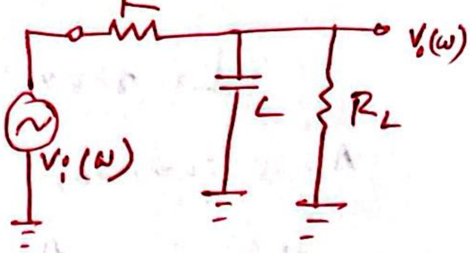


$$\therefore T(s) = \frac{1}{1+\omega_c/s} \quad s = j\omega$$

$$\boxed{\therefore T(s) = \frac{A_o}{1+\omega_c/s}} \quad A_o \rightarrow \text{max gain}$$

General form of TF for HP

Passive RC filter with load:



$$Z_p = R_L \parallel \frac{1}{j\omega C} \quad C$$

$$V_o = \frac{Z_p}{R+Z_p} V_i = \frac{1}{1+R/Z_p} V_i$$

$$T(j\omega) = \frac{V_o}{V_i} = \frac{1}{1+R/Z_p}$$

→ R_L input resistor of the amplifier or microphone.

$$= \frac{1}{1+R\left(\frac{1}{R_L}+j\omega C\right)}$$

$\frac{1}{Z_p} = Y_p$ = Admittance

$= \frac{1}{R_L} + j\omega C$

$$= \frac{1}{1+\frac{R}{R_L}+\cancel{\frac{R}{j\omega C}}\ Rj\omega C}$$

$$= \frac{1}{\frac{R+R_L}{R_L}+j\omega CR} \times \frac{R_L/(R+R_L)}{R_L/(R+R_L)}$$

$$= \frac{R_L/(R+R_L)}{1+j\omega CRR_L/(R+R_L)}$$

Now,

$R_L/(R+R_L) = A_o$ = constant

$RR_L/(R+R_L) = R \parallel R_L = R_p$ = parallel equivalent

$$\therefore T(j\omega) = \frac{V_o}{V_i} = \frac{A_o}{1+j\omega CR_p}$$

$$\therefore T(s) = \frac{A_o}{1+sCR_p} = \frac{A_o}{1+\omega_{cL}/s}$$

$$\omega_{cL} = \frac{1}{CR_p} > \omega_c \quad (\omega_c = 1/RC)$$

$R_p = R \parallel R_L < R$

↳ Cut off freq^n with load

$$A_o = \frac{R_L}{R+R_L} < 1$$

↳ max gain with load.

As a summary

$$\therefore \; T(s) = \frac{A_o}{1+\omega_{cL}/s}$$

$$A_o = \frac{R_L}{R+R_L} < 1$$

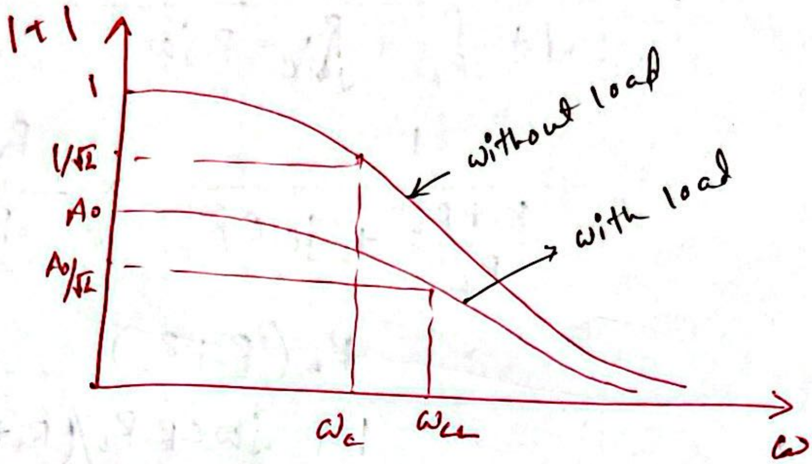
$$\omega_{cL} = \frac{1}{R_p C} > \omega_c$$

For RC Low pass without load

$$T(s) = \frac{1}{1+\omega_c/s}$$

$$\omega_c = \frac{1}{RC}$$

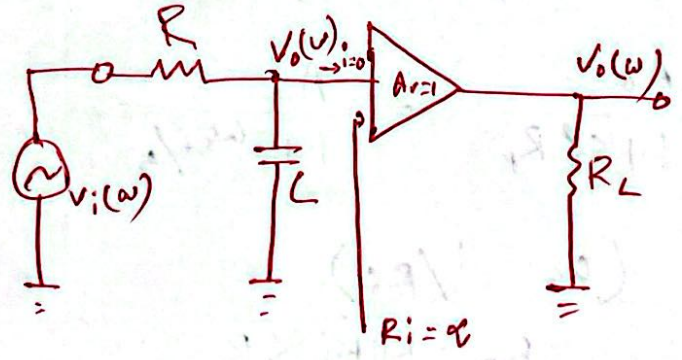
$$A_o = 1$$



Drawback

changes filter characteristics
→ unwanted signal introduction due to the increase in cut off freq
→ Lowers the gain of the system

Solⁿ



$$V_o = \frac{1/j\omega C}{R + 1/j\omega C} V_i$$

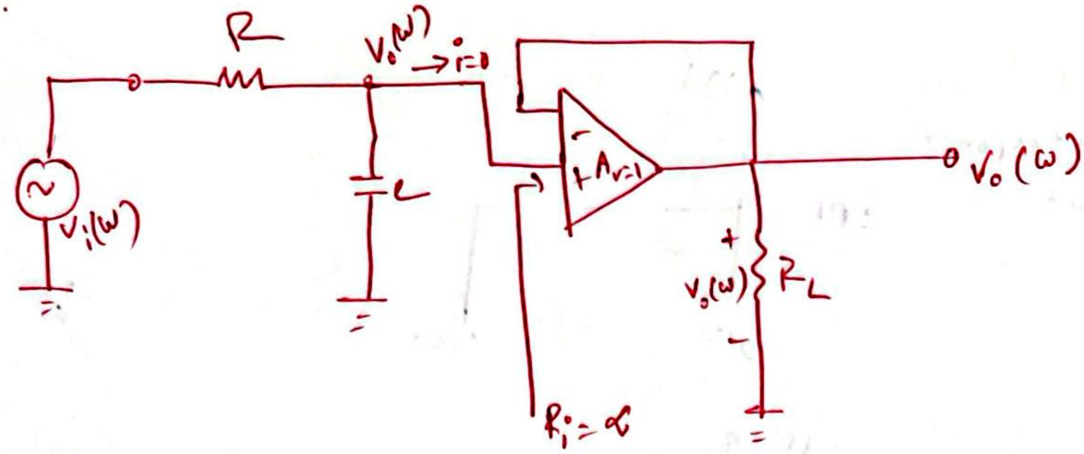
$$= \frac{1}{1+j\omega CR} V_i$$

$$T(j\omega) = \frac{V_o}{V_i} = \frac{1}{1+j\omega CR}$$

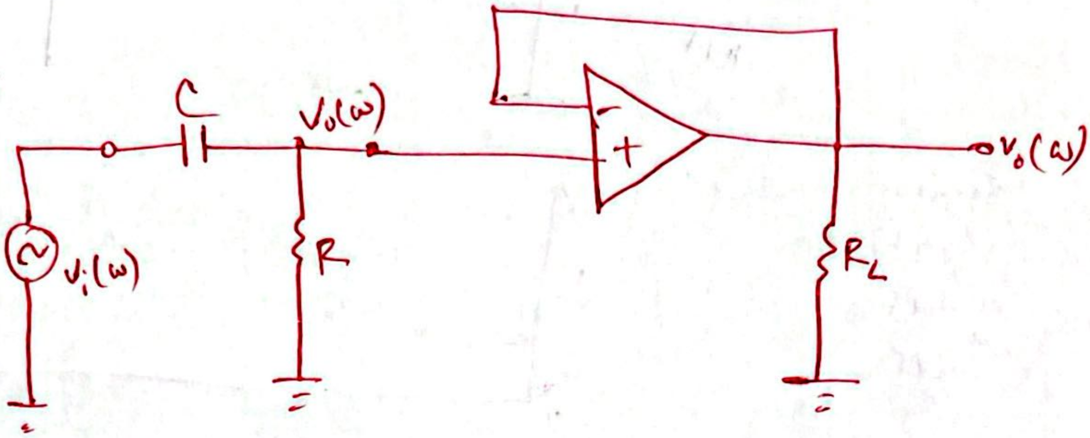
$$= \frac{1}{1+j\omega/\omega_c}$$

$$T(s) = \frac{1}{1+s/\omega_c}$$

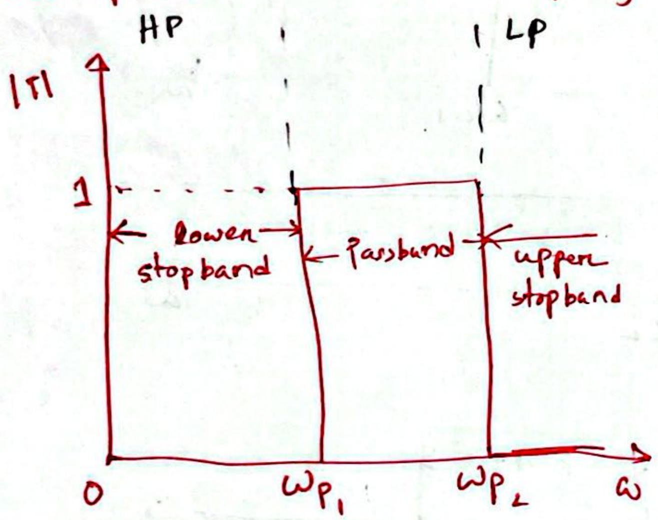
2/ Active LP:



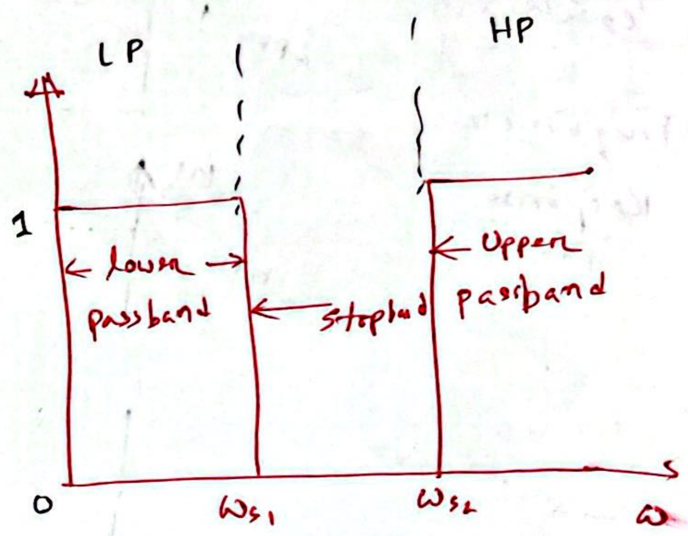
Active HP:



Bandpass and Bandstop filters:

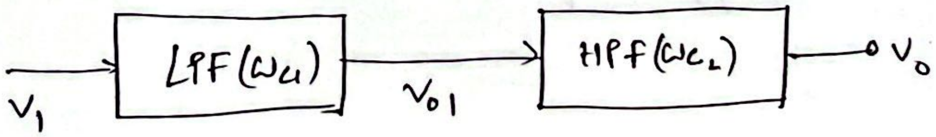


BPF



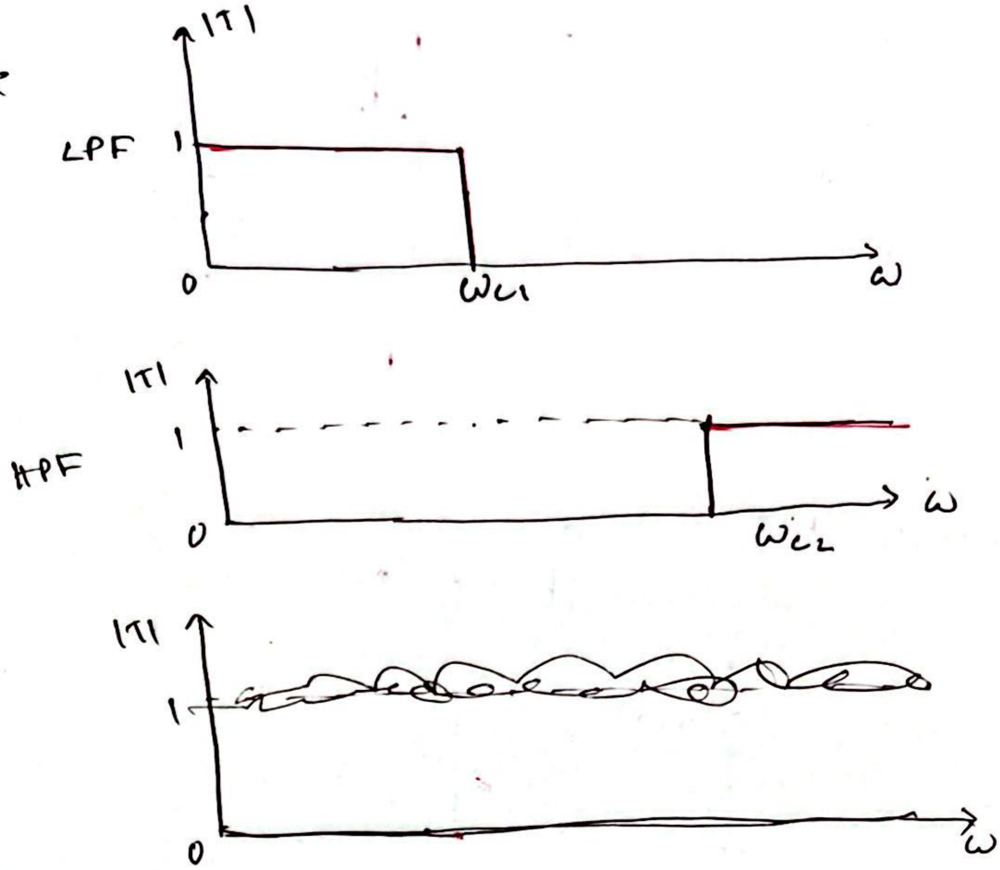
BS

Series Combination:



$\boxed{\omega_{c2} > \omega_{c1}}$

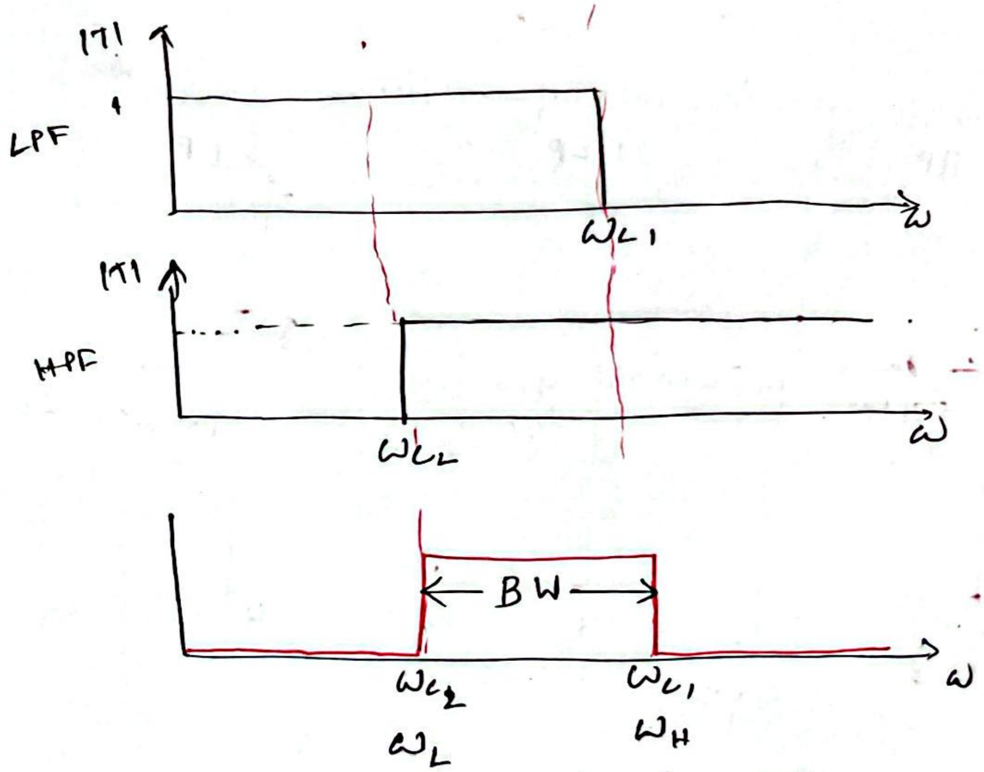
Frequency Response Plot



Series combination of HPF LPF

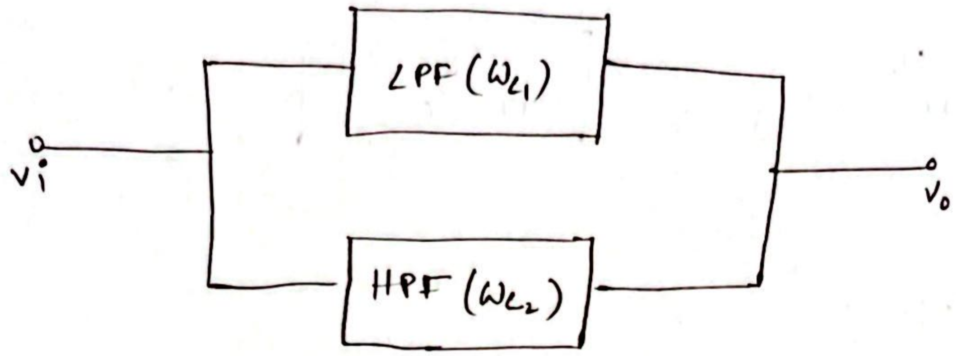
$\boxed{\omega_{c2} < \omega_{c1}}$

Frequency Response plot



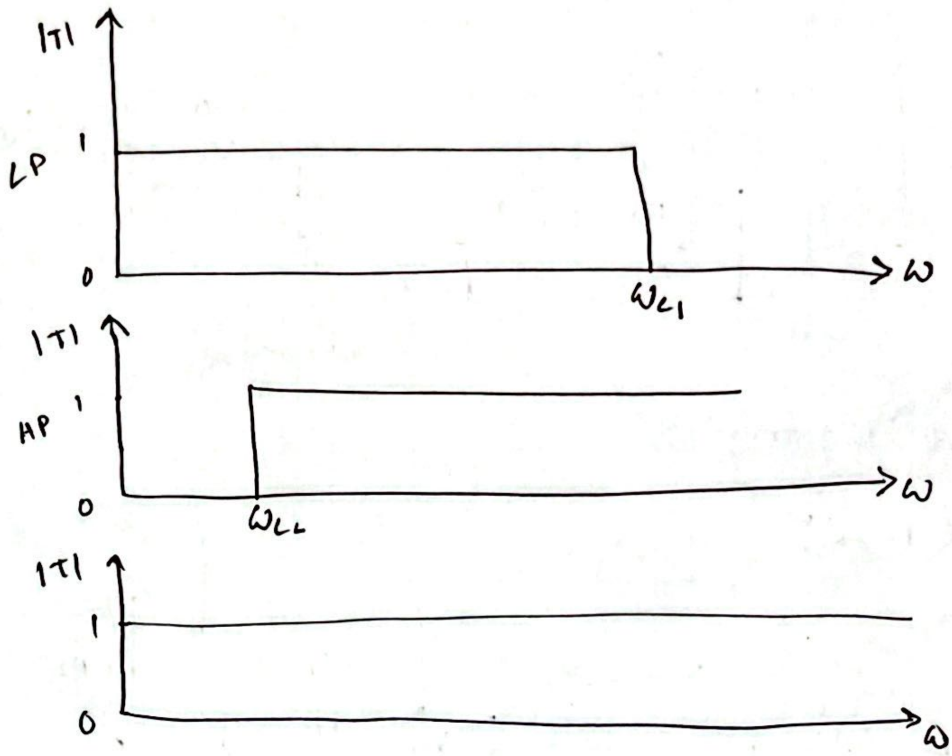
cutoff freq HP < cut off freq LP.

Parallel Combination:



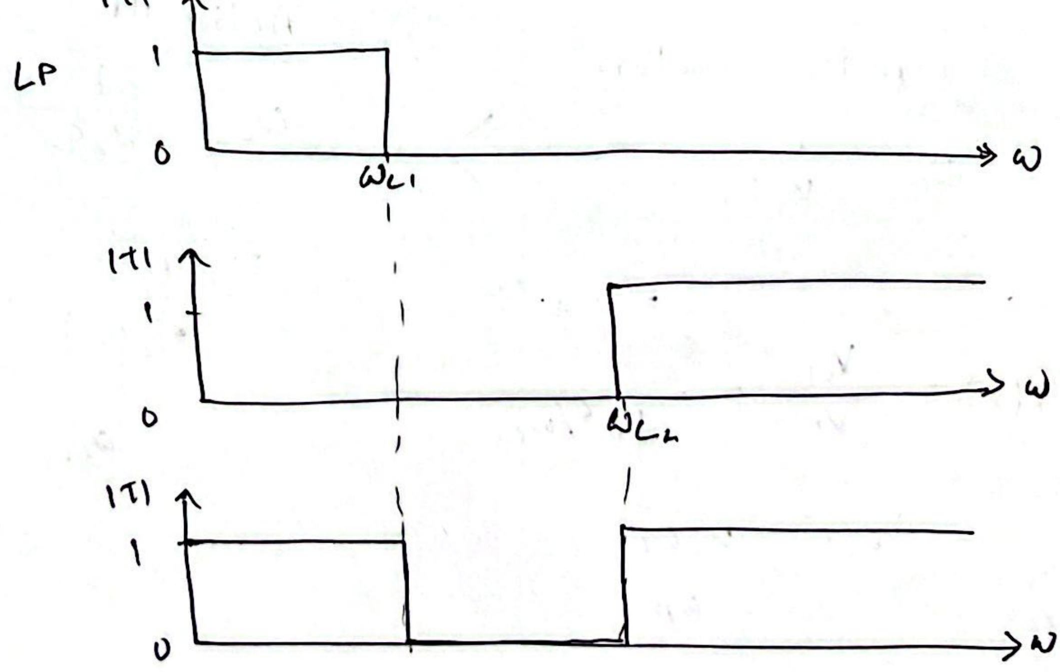
$\omega_{c_1} > \omega_{c_2}$

All Pass filter.

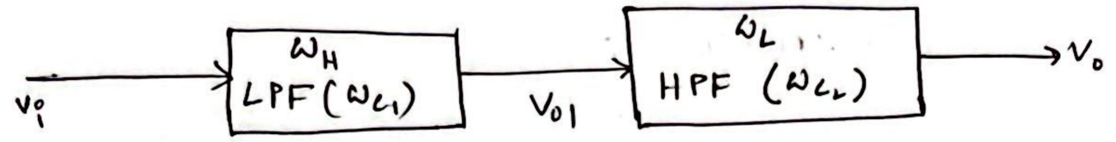


$\omega_{c_2} > \omega_{c_1}$

Band Stop filter.

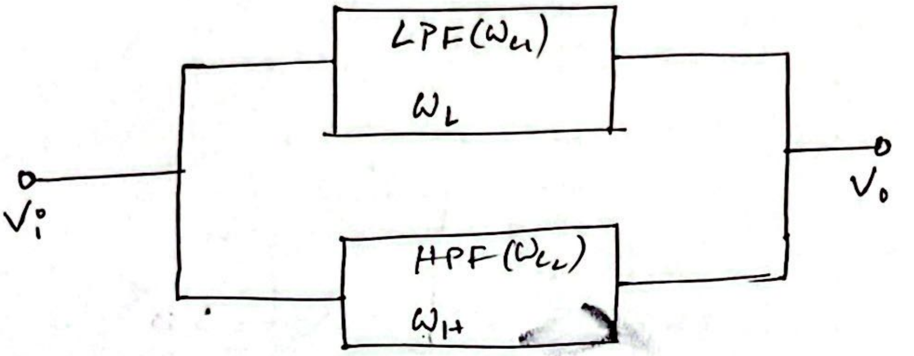


Band Pass filter:



$\omega_{c2} < \omega_{c1}$

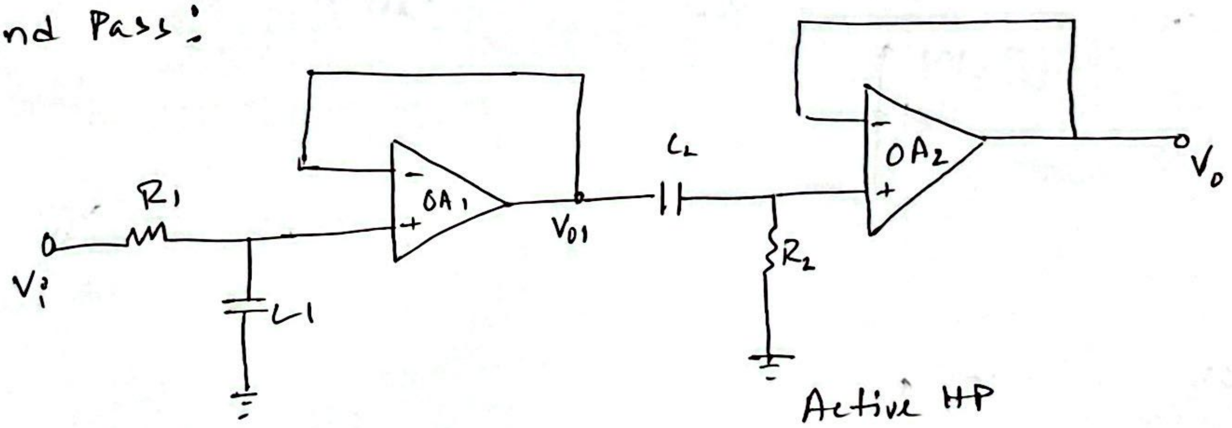
Band Stop filter:



$\omega_{c1} < \omega_{c2}$

Circuit Diagram:

Band Pass:



Active low Pass

$$\omega_{c1} = \frac{1}{R_1 C_1}$$

$>$

Active HP

$$\omega_{c2} = \frac{1}{R_2 C_2}$$

$= \omega_L$

$= \omega_H$

$$T_1(s) = \frac{V_{01}}{V_i} = \frac{1}{1 + s/\omega_{c1}}$$

$$\omega_{c1} = \frac{1}{R_1 C_1}$$

$= \omega_H$

$$T_2(s) = \frac{V_0}{V_{01}} = \frac{1}{1 + \omega_{c2}/s}$$

$$\omega_{c2} = \frac{1}{R_2 C_2}$$

$= \omega_L$

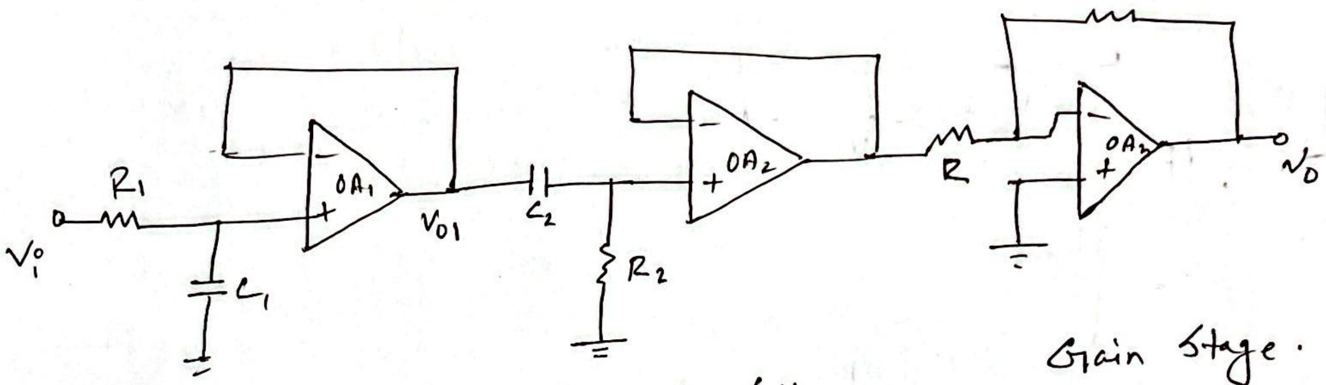
$\therefore$ Overall transfer function,

$$T(s) = \frac{V_o}{V_i} = \frac{V_o}{V_{o1}} * \frac{V_{o1}}{V_i}$$

$$= T_2(s) * T_1(s)$$

$$= \frac{1}{1+\omega_{c2}/s} * \frac{1}{1+s/\omega_{c1}}$$

Max gain = 1



Gain Stage.

$$T_3(s) = \frac{V_o}{V_{o2}}$$

$$= -\frac{R_f}{R}$$

HP filter

$$T_2(s) = \frac{V_{o2}}{V_{o1}}$$

$$= \frac{1}{1+\omega_{c2}/s}$$

LP filter

$$T_1(s) = \frac{V_{o1}}{V_i} = \frac{1}{1+s/\omega_{c1}}$$

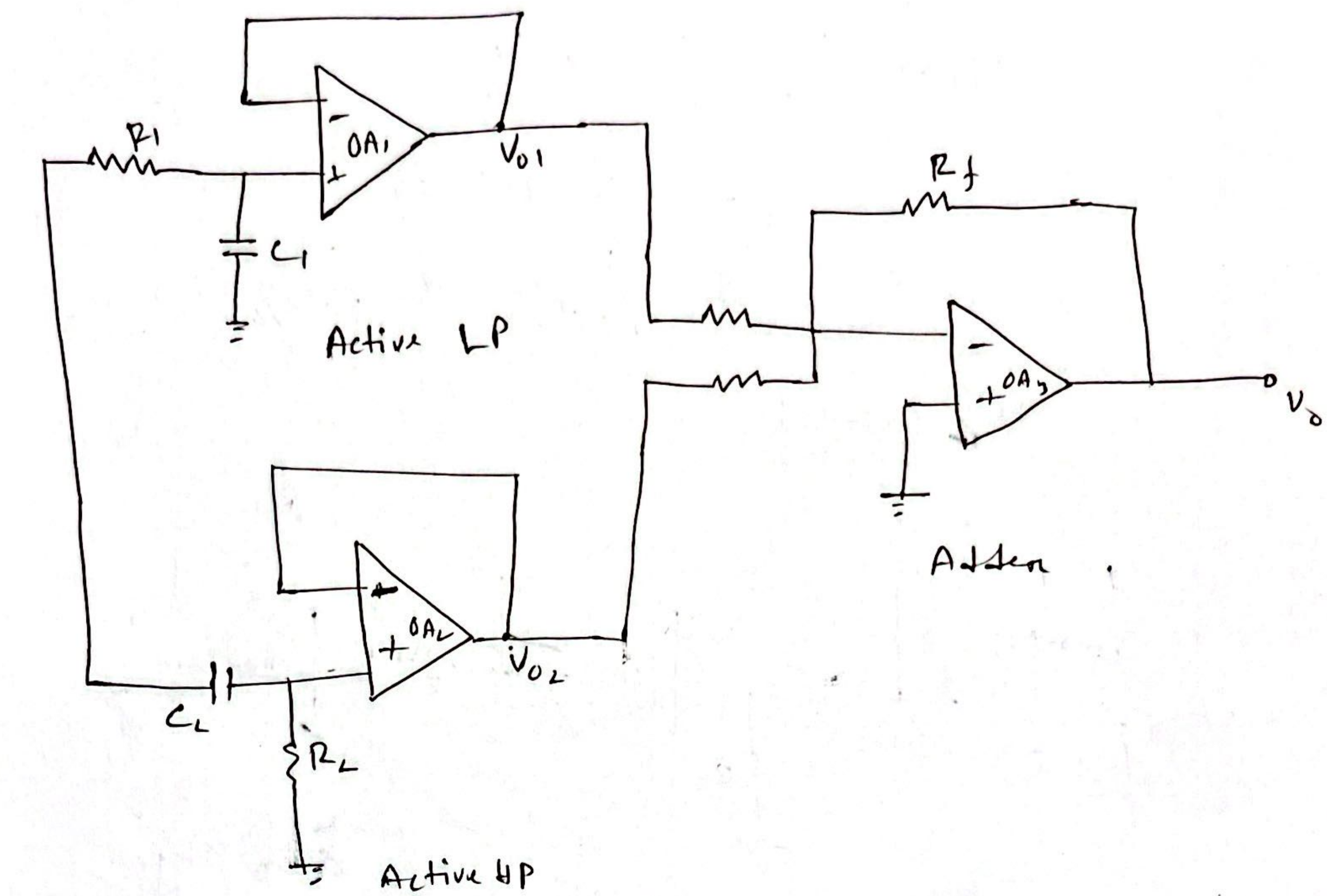
$\therefore$ Overall transfer function;

$$T(s) = \frac{V_o}{V_i} = \frac{V_o}{V_{o2}} * \frac{V_{o2}}{V_{o1}} * \frac{V_{o1}}{V_i}$$

$$= T_3(s) * T_2(s) * T_1(s)$$

$$= -\frac{R_f}{R} * \frac{1}{1+\omega_{c2}/s} * \frac{1}{1+s/\omega_{c1}}$$

Band Stop :



$$\omega_{c1} = \frac{1}{R_1 C_1} = \omega_L \qquad \omega_{c2} > \omega_{c1} \qquad \omega_{c2} = \frac{1}{R_2 C_2} = \omega_H$$

$$L.P \rightarrow T_1(s) = \frac{V_{o1}}{V_i} = \frac{1}{1 + s/\omega_{c1}}$$

$$H.P \rightarrow T_2(s) = \frac{V_{o2}}{V_i} = \frac{1}{1 + \omega_{c2}/s}$$

$\therefore$ Overall transfer function,

$$V_o = -\frac{R_f}{R}\left(V_{o1} + V_{o2}\right)$$

$$T(s) = \frac{V_o}{V_i} = -\frac{R_f}{R}\left(\frac{V_{o1}}{V_i} + \frac{V_{o2}}{V_i}\right)$$

$$= -\frac{R_f}{R}\left(T_1(s) + T_2(s)\right)$$

$$= -\frac{R_f}{R}\left(\frac{1}{1 + s/\omega_{c1}} + \frac{1}{1 + \omega_{c2}/s}\right)$$